

Retrieve Real-time Data in Android Using Firebase

Firebase is a cloud based platform for mobile and Web application development. In this article, which is aimed at Android enthusiasts, we explain how to use Firebase's cloud services to store and retrieve real-time data.



In today's digital economy, most of our interactions, such as streaming stock exchange news, weather reports or news updates, occur in near real-time. The need for real-time data is shaping the way technology and markets are evolving currently. While the mobility trend and Android usage have grown over the past decade, combining mobility with real-time data will open up a plethora of opportunities to explore. Firebase is one cloud based platform that has great support for dealing with real-time data when it comes to Android. In this article, I will explain how to integrate Firebase with your app in Android Studio, with an illustration.

For start-ups, Firebase is a quick and easy-to-use cloud based platform for making apps without worrying about servers, cloud storage and high availability among those servers.

About Firebase

Firebase is a unified mobile and Web app development platform from Google. It serves as a back-end as a service that provides real-time data syncing with all the clients subscribed to it, at a given instant. It also gives tools to rapidly develop high-quality apps with quick turnaround times. It is easy to implement Android apps using Firebase

without the need for any server side expertise or any knowledge of the database.

Firebase provides a real-time database for storing and syncing your app's data. The data is stored in JSON format and synchronised to every connected client. Besides, it automatically backs up user data every day without any performance or bandwidth impact on the app.

Getting started with Firebase

Before integrating your app with Firebase, the pre-requisites are:

1. Android Studio version 1.5 or higher, and the device should be running Android 2.3 (Gingerbread) or above. Android Studio being the official SDK, it can be downloaded from the Android developer's website.
2. A Firebase account is needed to be able to use its services. You can create a free account by visiting <https://www.firebase.com/> and signing up using your Google credentials.
3. SDK tools: Android studio needs to have Google Play services (version 9.2.0 or newer) installed. If not, you can install it using the SDK manager of Android Studio.